

Storage Management (Basic + LVM)

◇ Section 1: Basic Disk Partitioning and Mounting

```
# list the block devices
lsblk

# list all partitions
sudo fdisk -l

# manage partitions on disk
sudo fdisk /dev/sda

# create filesystem (ext4)
sudo mkfs.ext4 /dev/sda1

# create filesystem (xfs)
sudo mkfs.xfs /dev/sda2

# create mount points
sudo mkdir /mnt/dir1
sudo mkdir /mnt/dir2

# mount partitions
sudo mount /dev/sda1 /mnt/dir1
sudo mount /dev/sda2 /mnt/dir2

# check disk usage
df -h

# edit fstab for permanent mount
sudo vim /etc/fstab

# add following entries
# partition      mount point      fs      defaults      defaults
/dev/sda1        /mnt/dir1         ext4     defaults      0 0
/dev/sda2        /mnt/dir2         ext4     defaults      0 0
```

◇ Section 2: LVM Creation and Mounting

```
# list block devices
lsblk

# list partitions
sudo fdisk -l
```

```

# create partitions on disks
sudo fdisk /dev/sda
sudo fdisk /dev/sdb

# create physical volumes
sudo pvcreate /dev/sda1
sudo pvcreate /dev/sdb1

# display physical volumes
sudo pvdisplay

# create volume group
sudo vgcreate myvg /dev/sda1 /dev/sdb1

# display volume group
sudo vgdisplay

# create logical volumes
sudo lvcreate -n lv1 -L 1024M myvg
sudo lvcreate -n lv2 -L 2.5G myvg

# display logical volumes
sudo lvdisplay

# create filesystem on logical volumes
sudo mkfs -t ext4 /dev/myvg/lv1
sudo mkfs -t ext4 /dev/myvg/lv2

# mount logical volumes
sudo mount /dev/myvg/lv1 /mnt/dir1
sudo mount /dev/myvg/lv2 /mnt/dir2

# check mounted filesystems with type
df -Th

# edit fstab for permanent mount
sudo vim /etc/fstab

# add following entries
# partition      mount point    fs      defaults      defaults
/dev/mapper/myvg-lv1 /mnt/dir1    ext4    defaults      0 0
/dev/mapper/myvg-lv2 /mnt/dir2    ext4    defaults      0 0

```

◇ Section 3: Extending LVM

```

# list block devices
lsblk

# create partition on new disk
sudo fdisk /dev/sdc

```

```
# create physical volume
sudo pvcreate /dev/sdc1

# extend volume group
sudo vgextend myvg /dev/sdc1

# extend logical volume
sudo lvextend /dev/myvg/lv2 -L 4.5G

# resize filesystem (ext4)
sudo resize2fs /dev/myvg/lv2 4608M

# check updated disk usage
df -h
```

systemd services

```
# get the list of unit files managed by systemd
> sudo systemctl list-unit-files

# get the list of unit files of a selected type
# > sudo systemctl list-units --type=<type>
> sudo systemctl list-units --type=service
> sudo systemctl list-units --type=mount
```